8.3 Box-and-Whisker Plots

Name _____ Period ____ Date _____

Algebra I 8.3 · Student sheet

I can…

- find the five-number summary of a data set.
- make and read a box-and-whisker plot.
- use the range, interquartile range, and shape to describe and compare data sets.

Vocabulary · fill in the missing word

Quartiles — values that _____ a data set into four equal parts: Q1 is the median of the lower half, Q3 the median of the upper half

Five-number summary — least value, Q1, median, Q3, _____ value

Box-and-whisker plot — a display of the _____ summary; the box goes from Q1 to Q3 with a line at the median, and the whiskers reach the least and greatest values

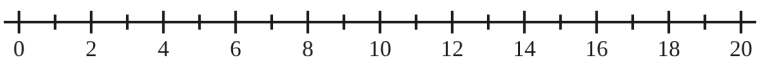
Interquartile range (IQR) — Q3 − Q1, the _____ of the middle half of the data

Part 1 · Human box plot

Record the numbers on the cards in order:

Least	Q1	Median	Q3	Greatest	Range	IQR

Sketch the box-and-whisker plot:



Part 2 · Notes: fill in the blanks

Five-number summary: _____, _____, _____, _____, _____

Q1 is the median of the _____ half; Q3 is the median of the _____ half.

IQR = _____ − _____, the spread of the middle _____ of the data.

A longer right whisker means the distribution is skewed _____.

Part 3 · Examples (work them with the class)

Example 1

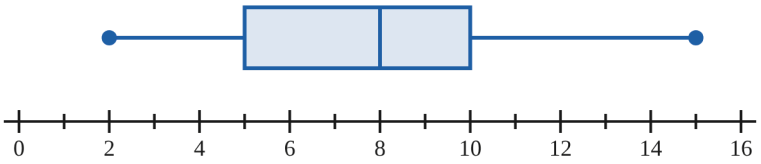
Ages of 10 people at a party: 18, 25, 21, 30, 19, 23, 27, 22, 20, 35. Find the five-number summary and make a box-and-whisker plot.

Example 2

Use the box-and-whisker plot.

a. Find the median. b. Find the range. c. Find the IQR.

d. About what percent of the data are between 5 and 10?

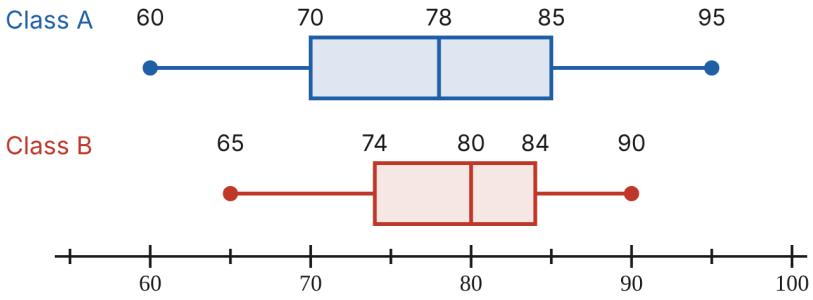


Example 3

The plots show the quiz scores of two classes.

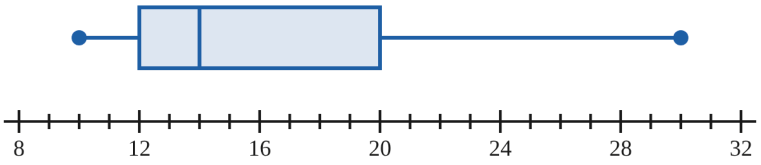
a. Which class has the greater median?

b. Which class's scores are more spread out? Use the range and IQR.



Example 4

Describe the shape of the distribution.



Part 4 · Partner practice

Partner 1

Find the five-number summary and the IQR: 7, 15, 9, 12, 20, 11, 14, 8, 16

Partner 2

Find the range and the IQR. Are the data more spread out below Q1 or above Q3?

